Extracting Software Mentions and Relations using Transformers and LLM-Generated Synthetic Data at SOMD 2025

Leveraging Transformers for NER and Relation Extraction Using LLM-Generated Synthetic Data

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Problem Statement

Software has become an indispensable component of modern scientific research, serving not only as a tool for analysis and experimentation but also as a topic of theoretical interest. However, despite its ubiquity and importance, software is frequently under-cited or informally referenced in scholarly publications. This lack of systematic and standardized citation hinders reproducibility, traceability, and recognition of software contributions in research.

The SOMD 2025 shared task aims to address this challenge by developing automated systems for software mention detection in scientific literature. The core objectives include identifying software-related entities—such as software names, versions, developers, licenses, and usage contexts—and extracting semantic relationships between these entities, including versioning and plugin dependencies.

This task is framed as a combination of Named Entity Recognition (NER) and Relation Extraction (RE) problems. NER involves detecting fine-grained software-related entities using a hierarchically annotated dataset with a BIO tagging scheme, while RE involves classifying the relations between entity pairs. Addressing this problem effectively requires overcoming challenges such as varied and ambiguous software naming conventions, complex contextual usage, and limited annotated data, motivating the use of advanced transformer models and synthetic data generation via instruction-tuned large language models.

Background

Software plays a fundamental role in contemporary scientific research—enabling experimentation, simulation, analysis, and ensuring reproducibility. However, despite its centrality, software is often informally mentioned in scholarly articles and rarely receives proper citation or metadata annotation. This lack of standardization hinders effective indexing, reuse, and reproducibility of research (Schindler et al., 2021).

The annotated dataset for this task presents unique challenges, including class imbalance and long-tail distributions. Over 82% of tokens are non-entities ("O"), and the majority of relation instances (85%) are concentrated in just five categories—Version of, Developer of, Citation of, URL of, and PlugIn of. This imbalance necessitates robust modeling techniques capable of handling sparse and diverse label distributions effectively.

Training Model Architecture

Named Entity Recognition (NER)

To tackle the fine-grained and imbalanced nature of software-related entity detection, we designed a scalable pipeline using **DeBERTa-v3-large** (He et al., 2021), an encoder-only transformer known for its disentangled attention and robust contextual representations. A token classification head is added on top, trained with a **BIO tagging scheme** to identify software-related entities.

Given that over **80%** of tokens are non-entities and that true entity mentions are sparse and context-dependent, the model requires sensitivity to nuanced patterns. To address these challenges, we adopted several training strategies.

These include **warmup scheduling** and **weight decay** for regularization, **gradient accumulation** to handle memory constraints, and **negative sampling** with a downsampling ratio of **0.3** to mitigate class imbalance and enhance the model's focus on minority entity classes.

Relation Extraction (RE)

For RE, we formulated the task as a **multi-class classification** over pairs of identified entities. Our context-aware strategy involved concatenating the full input sentence with entity spans and their associated types, enabling the model to leverage both sentence-level semantics and entity-level details.

We experimented with **DeBERTa-v3-large** and **ModernBERT** (Warner et al., 2024) as encoders. The architecture employs **mean pooling** over token embeddings (weighted by the attention mask), followed by a **linear classification head** that projects to **12 relation types**.

Dropout was set to **0.1** for both hidden and attention layers to prevent overfitting. The model was trained efficiently on **dual T4 GPUs** using Kaggle's public infrastructure. Despite this robust setup, RE remains a challenging task due to class imbalance and the subtle nature of relation semantics, resulting in a **macro-averaged F1 score of approximately 15%**.

Data Augmentation Strategies

Example of Synthetically Generated Data

Task: Merge and rephrase the following two texts at both syntactic and semantic levels while preserving specific tokens.

Text 1: *Significance was accepted at P ≤ 0.05, statistical power was 90%, and the analyses were performed in Statistical Package for Social Sciences version 20.0 (SPSS, Chicago, Illinois, USA).*

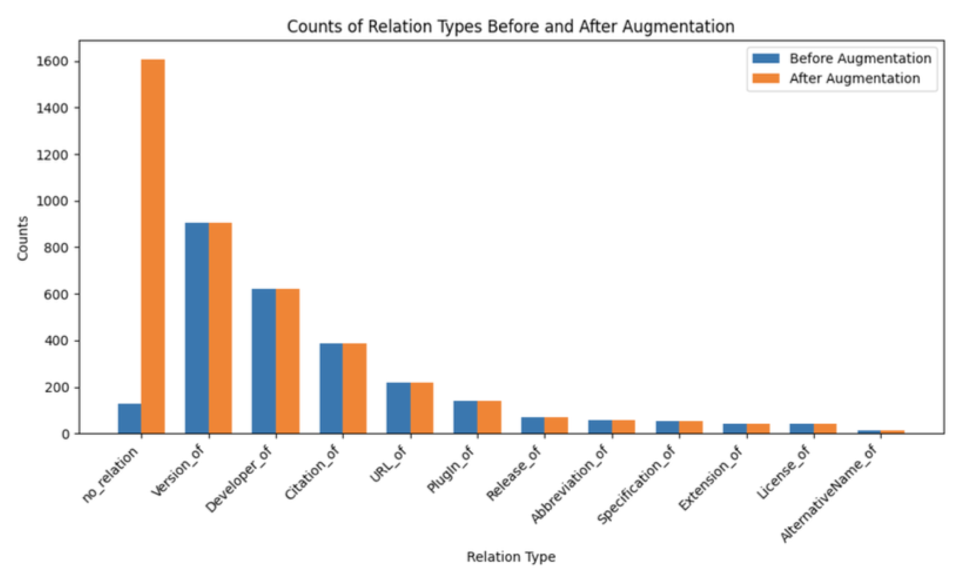
Text 2: *Microsoft Excel 2007 was used to compile the data and STATA Version 12.0 was used for statistical analyses [28].*

Required Tokens (to be preserved as-is): {`Microsoft`, `Excel`, `2007`, `12.0`, `[28]`, `Statistical`, `Package`, `for`, `Social`, `Sciences`, `20.0`, `SPSS`}

Generated Output Text:
In order to analyze the data, statistical significance was accepted at P ≤ 0.05 and the statistical power was determined to be 90%. The analyses were performed using **Statistical Package** for **Social Sciences** version **20.0**, commonly referred to as **SPSS**. Data compilation was carried out using **Microsoft Excel 2007**, while further statistical analysis was conducted using STATA version **12.0** ([28]).

We employed targeted data augmentation methods for both Relation Extraction (RE) and Named Entity Recognition (NER) to enhance model performance and address data sparsity.

1. **Named Entity Recognition (NER) Using LLMs**
   - Large Language Models (LLMs) are prompted to generate synthetic sentences that preserve original entities while varying the syntax.
   - Two training samples are paraphrased and merged, keeping all original entities intact.
   - The prompts include entity maps and token constraints to ensure perfect label recovery.
   - Used models: **Gemma-2-9b-it** (Gemma Team et al., 2024), **Mistral-7b-instruct-v0.1** (Jiang et al., 2023), and **Qwen2.5-7b-instruct** (Yang et al., 2024)(Team, 2024).
   - Generated text retains alignment of the entity span while introducing richer linguistic diversity.
2. **Relation Extraction (RE)**
   - Each sentence is paired with two entities formatted as: `entity_type [SEP] entity_text`.
   - Annotated entity pairs are labeled with true relations such as `Developer_of` or `URL_of`.
   - *Hard negatives* are created by forming new pairs of non-annotated entities and labeling them as `no_relation`..
   - Input format: `[document text] [SEP] entity_1 [SEP] entity_2`.
   - This design helps the model differentiate true relations from spurious co-occurrences.



These strategies increase the robustness of the dataset and generalizability of the model by introducing harder negatives and broader syntactic contexts.

Results

Named Entity Recognition (NER)

We evaluated several modeling strategies for NER, focusing on **DeBERTa-v3-large** with a token classification head.

- Applied **4-fold cross-validation**; best fold (Fold 1) achieved a macro F1 score of **60%**.
- Incorporated **synthetic training data** from instruction-tuned LLMs to boost generalization:
  - **Mistral-7B-Instruct-v0.1** yielded a **+5%** improvement in macro F1.
  - **Qwen2.5-7B-Instruct** and **Gemma-2-9B-it** offered a modest **+2%** gain.
- **Best setup:** DeBERTa-v3-large + Mistral data → **65%** macro F1 on test set.
- XLM-RoBERTa underperformed with only **28%** macro F1.

Task	Model / Setup	Precision	Recall	F1
NER	Deberta-V3-Large	0.5734	0.6612	0.5993
	Deberta-V3-Large (Full Fit + Mistral-7B)	0.6482	0.6943	**0.6498**
	Deberta-V3-Large (Full Fit + Gemma2-9B)	0.5875	0.6808	0.6199
	Deberta-V3-Large (Full Fit + Qwen2.5)	0.6657	0.6531	0.6215
	XLM-RoBERTa (Full Fit + Gemma2-9B)	0.2775	0.3104	0.2871
RE	Deberta-V3-Large	0.1025	0.4117	0.1543
	Modern BERT-Large	0.0878	0.4228	0.1379
	Deberta-V3-Large (Augmented Data)	0.3785	0.6744	**0.4675**
	Modern BERT-Large (Augmented Data)	0.3473	0.6702	0.4384

Relation Extraction (RE)

Initial RE models showed overfitting to dominant classes and spurious entity correlations.

- Introduced **hard negative augmentation** by adding entity pairs with `no_relation` labels.
- This balancing improved model robustness and overall performance.
- **Final macro F1 score: 47%**, a substantial gain of **+32%**.
- The final RE model, trained on the full dataset with augmentation, was evaluated during the open submission phase.

Conclusion

Our approach achieves promising results for software-related Named Entity Recognition (NER) and Relation Extraction (RE) tasks. The NER model attains a macro F1 score of **65%**, with some entity categories like *Release* and *SoftwareCoreference* achieving perfect precision and recall. However, certain sparse or ambiguous categories such as *PlugIn* and *Extension* remain challenging.

Initial RE performance was limited by overfitting and misclassification, resulting in a macro F1 score of about **15%**. This was substantially improved to **47%** via hard negative data augmentation, demonstrating the importance of balanced training data for generalization.

Additionally, synthetic data augmentation with LLMs particularly Mistral-7B provided a notable performance boost of **5%** in F1 score. Our best model, DeBERTa-v3-large trained with augmented data, achieves an overall combined macro F1 score of **56%**.

These findings highlight the potential of integrating pretrained transformers with synthetic data to address complex information extraction challenges in scholarly text.

Future Work

Future efforts will focus on improving detection and classification for sparse and ambiguous entity categories, such as *PlugIn* and *Extension*. Further exploration of more diverse and larger-scale synthetic data generation using LLMs could enhance model generalization.

Additionally, refining relation extraction techniques to better handle imbalanced classes and more subtle semantic relationships remains a priority. Incorporating joint modeling approaches that simultaneously optimize NER and RE could reduce error propagation and improve overall system robustness.

We aim to extend these methods toward more comprehensive software mention detection systems that better support reproducibility, indexing, and citation practices in scientific literature.